

PRINCE SAINI

AI / ML Research | Deep Learning & Computer Vision | Multimodal AI | Remote Sensing | PyTorch
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RESEARCH SUMMARY

AI/ML researcher specializing in Deep Learning, Computer Vision, and Multimodal AI for geospatial and satellite-scale systems. Designs and trains PyTorch-based CNN and Vision Transformer architectures for remote sensing, sensor fusion, and image super-resolution, with applied research in transformer-based language models (GPT, BERT, LLaMA) and LLM fine-tuning. Builds end-to-end research pipelines spanning multimodal data preprocessing, feature engineering, model training, evaluation, and deployment-ready workflows. Targeting a research-focused internship advancing deep learning systems at production scale.

TECHNICAL SKILLS

Deep Learning: PyTorch, CNNs, Vision Transformers (ViTs), Transformer Architectures (GPT, BERT, LLaMA), LLM Fine-Tuning, Generative AI

Computer Vision: OpenCV, Image Processing, Satellite Image Analysis, Multimodal Data Fusion

Machine Learning: XGBoost, Random Forest, Scikit-learn, Feature Engineering, Model Evaluation & Benchmarking

Remote Sensing / Geospatial: Geospatial Analytics (GIS), QGIS, SeaDAS, Satellite Imagery Analysis

Programming & Tools: Python, C++, Java, SQL, NumPy, Pandas, Git/GitHub, Linux, Jupyter

RESEARCH EXPERIENCE

Research Intern — Deep Learning & Remote Sensing, INCOIS

Conduct applied deep learning research on multispectral and thermal satellite imagery for maritime surveillance and ocean monitoring, engineering end-to-end AI pipelines across three independent research initiatives.

Automated Dark Vessel Detection via Multimodal Analytics *Python | PyTorch | CV | Geospatial AI | [GitHub](#)*

- Engineered an end-to-end multimodal AI pipeline fusing VIIRS-DNB, Sentinel-1 SAR, AIS, and thermal satellite imagery to detect vessels evading AIS transmission.
- Automated preprocessing, candidate extraction, feature engineering, and anomaly-detection workflows across heterogeneous geospatial data sources.
- Trained a vessel risk-prediction model achieving 96.43% classification accuracy, powering automated geospatial reporting for maritime intelligence.

AI-Based Super-Resolution of Sea Surface Temperature (Ongoing) *Python | PyTorch | CNN*

- Architected a CNN-based super-resolution pipeline to downscale 1 km satellite SST data to 10 m, reconstructing fine-scale thermal gradients from coarse observations and validating fidelity against higher-resolution reference imagery.

Marine Plastic Detection Using EOS-06 OCM Satellite Data *Python | DL | Remote Sensing | [GitHub](#)*

- Built an AI-assisted detection framework for marine plastic accumulation from EOS-06 multispectral imagery; benchmarked CNN, XGBoost, and Random Forest on radiometrically corrected data and characterized spectral signatures of floating debris to inform scalable pollution-monitoring workflows.

ADDITIONAL PROJECT

Gesture-Driven Smart Control System *Python | Embedded AI | BLE | IoT*

- Engineered a wearable gesture-recognition system (Arduino Nano 33 BLE Sense + MPU6050) with a Flutter cross-platform app, achieving 94.8% classification accuracy in real time.
- Validated robustness via a 700-trial Gaussian noise-injection simulation across 7 gesture classes; architected dual BLE/MQTT pathways achieving sub-500ms end-to-end latency (~290-320ms BLE, ~450ms MQTT).

PUBLICATIONS

- Automated Dark Vessel Detection via Multimodal Analytics: Integrating VIIRS-DNB, SAR, Thermal & AIS Data (manuscript in preparation)
- Gesture-Driven Smart Control System for Intelligent Human-Computer Interaction (manuscript in preparation)

CERTIFICATIONS

- IBM AI Engineering Professional Certificate
- IBM Generative AI Engineering Professional Certificate
- AWS Machine Learning Engineer – Associate

EDUCATION

Vellore Institute of Technology (VIT), Bhopal — B.Tech, Computer Science and Engineering *Expected 2027*

CGPA: 7.5 / 10